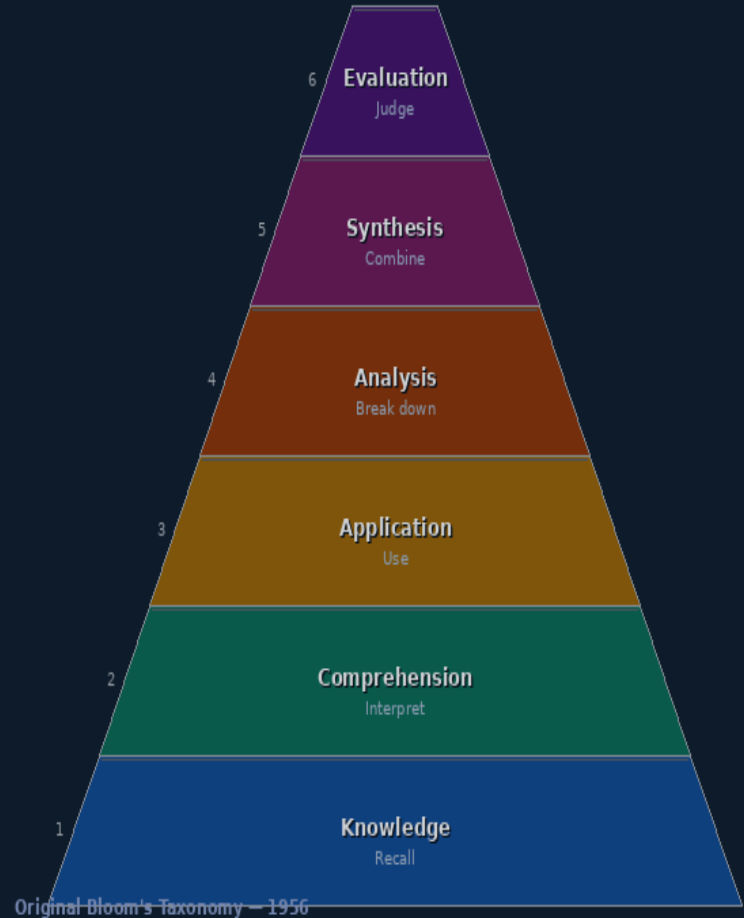


BLOOM'S

Taxonomy

A Comprehensive Lecture

- Original vs. Revised Taxonomy
- Three Domains of Learning
- Verbs & Practical Examples
- Formulating Goals & Objectives
- The ABCD Model



Benjamin S. Bloom & the Origins

How a 1948 committee meeting changed educational theory forever



Benjamin S. Bloom

1913 – 1999

Psychologist & Educator
University of Chicago

1948

American Psychological Association (APA) conference — Bloom proposes a classification system for educational objectives.

1956

Taxonomy of Educational Objectives: Cognitive Domain published — the original 6-level hierarchy revolutionizes curriculum design.

1964

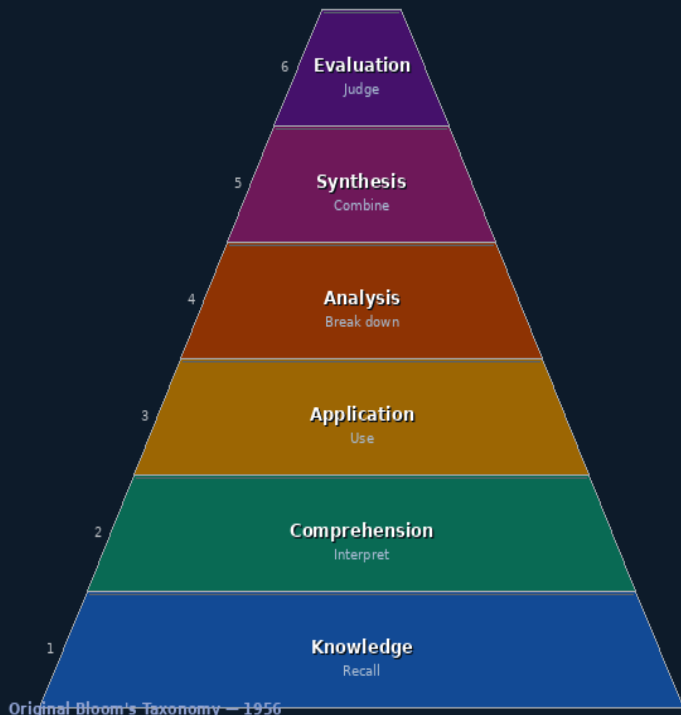
Krathwohl, Bloom & Masia publish the Affective Domain taxonomy, addressing attitudes and values.

2001

Anderson & Krathwohl revise the taxonomy — renaming levels with verbs, moving Synthesis→Create to the top.

The Original Taxonomy — 1956

Six levels in the Cognitive Domain — arranged from lower to higher order thinking



1

Knowledge

Recall of facts, terms, and basic concepts

e.g. List the 5 stages of cell division

2

Comprehension

Explain or interpret meaning from material

e.g. Explain how photosynthesis works

3

Application

Use information in new and concrete situations

e.g. Solve a chemistry equation

4

Analysis

Break material into parts; find patterns

e.g. Compare the causes of WWI vs WWII

5

Synthesis

Combine elements to form a new whole

e.g. Design an experiment to test a hypothesis

6

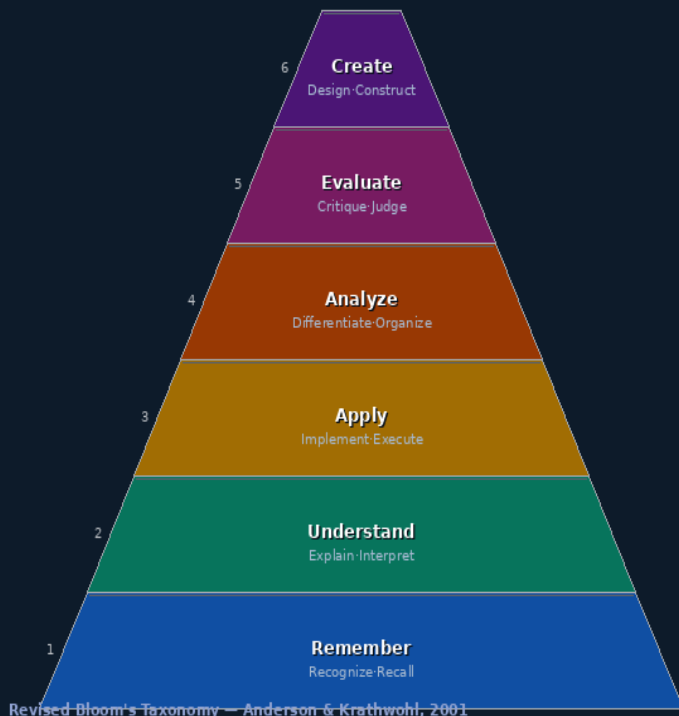
Evaluation

Judge value; justify decisions or courses of action

e.g. Assess the validity of a research paper

The Revised Taxonomy — 2001

Anderson & Krathwohl — nouns replaced by verbs, Synthesis renamed Create & moved to the apex



1

Remember

Retrieve relevant knowledge from memory

e.g. Name the bones of the human hand

2

Understand

Construct meaning from instructional messages

e.g. Summarize a research article in own words

3

Apply

Carry out a procedure in a given situation

e.g. Calculate medication dosage for a patient

4

Analyze

Break material into component parts; determine how they interrelate

e.g. Differentiate between Type 1 and Type 2 diabetes

5

Evaluate

Make judgments based on criteria and standards

e.g. Critique the methodology of a clinical trial

6

Create

Put elements together to form a novel whole

e.g. Design a patient care protocol from scratch

Original vs. Revised: What Changed?

The 2001 revision transformed nouns into action verbs and restructured the apex



Original (1956)	Revised (2001)	Key Change
Knowledge	Remember	Shifted focus from noun (content) to verb (action)
Comprehension	Understand	More active — learners construct meaning
Application	Apply	Verb form clarifies observable behavior
Analysis	Analyze	Consistent verb convention

Three Domains of Learning

Bloom identified that all learning occurs across three interconnected dimensions



Bloom's Three Domains of Learning

THINKING

Cognitive

Bloom, 1956 / Anderson & Krathwohl 2001 · 6 Levels

FEELING

Affective

Krathwohl, Bloom & Masia, 1964 · 5 Levels

DOING

Psychomotor

Simpson, 1972 · 5 Levels

Cognitive Domain: Verbs & Examples

Mental skills — from simple recall to complex creation. Use these verbs when writing learning objectives.

L1

Remember

Verbs: Define, List, Recall, Name, Identify, Label, Repeat, State

Example: List the 4 blood types

L2

Understand

Verbs: Explain, Summarize, Paraphrase, Classify, Describe, Interpret, Compare

Example: Explain the water cycle

L3

Apply

Verbs: Solve, Demonstrate, Calculate, Use, Execute, Implement, Operate

Example: Calculate medication dosage

L4

Analyze

Verbs: Compare, Differentiate, Examine, Break down, Distinguish, Organize

Example: Compare WWI & WWII causes

L5

Evaluate

Verbs: Judge, Justify, Critique, Defend, Assess, Argue, Prioritize

Example: Critique a research methodology

L6

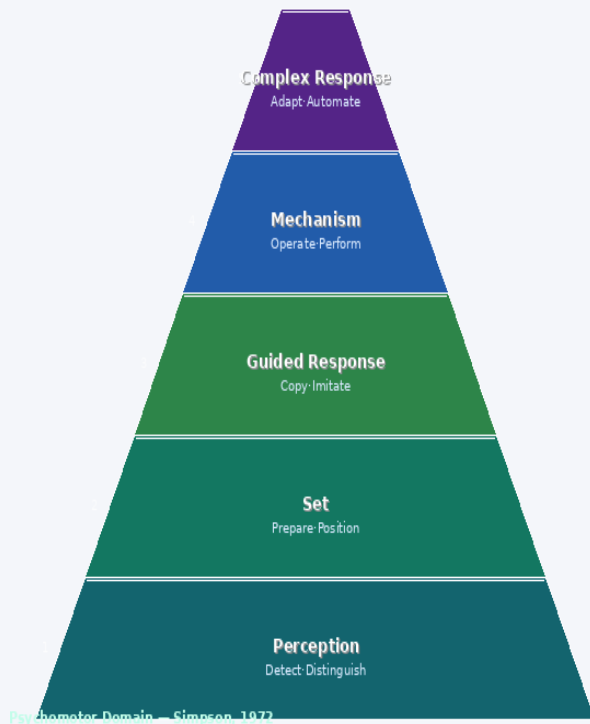
Create

Verbs: Design, Construct, Formulate, Develop, Compose, Plan, Produce

Example: Design a new patient care protocol

Psychomotor Domain: Verbs & Examples

Physical skills — motor coordination from imitation to mastery (Simpson, 1972)



L1

Perception

Verbs: Detect, Choose, Identify, Distinguish

Ex: Student notices incorrect grip on surgical tool

L2

Set

Verbs: Prepare, Position, Display, Begin, React

Ex: Positions hands correctly before starting IV insertion

L3

Guided Response

Verbs: Copy, Follow, Imitate, Attempt, Trace

Ex: Follows step-by-step guide to suture a wound

L4

Mechanism

Verbs: Calibrate, Construct, Operate, Perform

Ex: Operates ECG machine without supervision

L5

Complex Response

Verbs: Execute, Adapt, Coordinate, Refine, Automate

Ex: Performs laparoscopic procedure with full fluency

Affective Domain: Verbs & Examples

Attitudes, values & emotions — from passive awareness to internalized belief (Krathwohl et al., 1964)



L1

Receiving

Verbs: Listen, Attend, Notice, Accept, Be aware

Ex: Student listens attentively to a patient's concerns

L2

Responding

Verbs: Participate, Comply, Discuss, Contribute, React

Ex: Actively participates in ethics seminar discussion

L3

Valuing

Verbs: Appreciate, Prefer, Justify, Support, Advocate

Ex: Advocates for patient rights in team meetings

L4

Organization

Verbs: Organize, Prioritize, Integrate, Balance, Compare

Ex: Reconciles personal beliefs with professional ethics

L5

Characterization

Verbs: Internalize, Embody, Demonstrate, Revise, Develop

Ex: Consistently embodies patient-centered care values

Formulating Goals & Objectives

Understanding the critical distinction between broad goals and precise, measurable objectives

GOALS

Broad Intent

- General, long-term statements of educational purpose
- NOT directly measurable or observable
- Express ideals, aspirations, and values
- Apply to a course, program, or institution
- Guide the overall direction of instruction

Example: "Students will develop an appreciation for evidence-based practice."

OBJECTIVES

Specific Intent

- Specific, measurable statements of what learners DO
- Observable behaviors — can be seen or evaluated
- Include a clear standard or criterion for success
- Apply at the lesson, unit, or session level
- Drive assessment design and teaching strategies

Example: "Given a patient case, students will calculate oral medication dose with 90% accuracy."

Characteristics of Effective Objectives

What separates a well-written objective from a vague one?

S

Specific

Clearly states what the learner will DO. Uses action verbs, not vague terms like 'know' or 'understand'.

NOT: 'Understand anatomy'
YES: 'Label the 12 cranial nerves'

M

Measurable

Can be observed, counted, or evaluated objectively with a defined criterion.

NOT: 'Improve communication'
YES: 'Score ≥85% on OSCE station'

A

Achievable

Realistic for the learner's current level, prior knowledge, and available time.

NOT: Set for first-day students
YES: Built on prior foundational skills

R

Relevant

Aligned with the course goal, real-world needs, and the learner's future practice.

NOT: Memorize historical trivia
YES: Apply evidence-based protocols

T

Time-bound

Achievable within a defined instructional period — a session, unit, or semester.

NOT: 'Eventually be able to...'
YES: 'By end of Week 3, students will...'

The ABCD Model for Writing Objectives

Four essential components that make a learning objective complete and unambiguous

A

Audience

WHO are the learners?

Identify the specific group being taught. The objective describes what **THEY** will do, not what the teacher will teach. Always make the learner the subject.

- Nursing students
- Medical interns
- Grade 6 pupils

B

Behavior

What will they DO?

Use an observable **ACTION VERB** from Bloom's Taxonomy. Must be something that can be seen, heard, or evaluated. Avoid vague verbs like 'know', 'understand', or 'appreciate'.

- Identify
- Calculate
- Construct

C

Condition

Under what CIRCUMSTANCES?

Specify the context, tools, materials, or constraints given to the learner when performing the task.

- Given a patient case study...
- Without reference materials...
- Using a standard calculator...

D

Degree

How WELL must they perform?

Define the minimum acceptable standard — the criterion that separates success from failure. This makes the objective measurable.

- ...with 80% accuracy
- ...in 4 out of 5 cases
- ...in under 3 minutes

ABCD in Practice: Worked Examples

Three complete objectives across different disciplines — each component labeled

Healthcare / Nursing

- | | | | |
|----------|---|----------|--|
| A | First-year nursing students | B | will correctly measure and document vital signs |
| C | given a standardized simulated patient scenario | D | with 90% accuracy across 5 consecutive assessments |

Science Education

- | | | | |
|----------|---------------------------------------|----------|--|
| A | Grade 10 biology students | B | will construct and label a detailed diagram of cell division |
| C | without the use of textbooks or notes | D | correctly identifying all 5 phases with ≥80% accuracy |

Pharmacy Training

- | | | | |
|----------|---|----------|---|
| A | Third-year pharmacy students | B | will calculate and verify the appropriate drug dosage |
| C | given a laboratory practical exam with 10 patient scenarios | D | achieving a passing score of 9 out of 10 cases |



A = Audience



B = Behavior



C = Condition



D = Degree

Weak vs. Strong Objectives

Spot the difference — and fix it. This is the most practical skill in instructional design.

WEAK

"Students will know about infection control."



STRONG

Psychomotor —
Mechanism

"Given a clinical simulation, nursing students will demonstrate correct hand hygiene technique before and after patient contact in 5 out of 5 observations."

WEAK

"Learners will understand cardiovascular disease."



STRONG

Cognitive — Analyze

"After the cardiology lecture, students will compare the pathophysiology of atherosclerosis vs. hypertension using a Venn diagram with at least 4 correct distinctions."

WEAK

"Students will appreciate ethical practice."



STRONG

Affective — Valuing

"Following the bioethics workshop, students will defend their chosen position on a given case study in a written reflection of 300+ words, citing at least 2 ethical principles."

ABCD + SMART: Two Frameworks, One Objective

ABCD gives you the structure; SMART gives you the quality check. Use both together.

COMPLETE OBJECTIVE:

- | | |
|--|---|
| A Second-year pharmacy students | B will correctly calculate drug dosage |
| C given a practical exam with patient scenarios | D achieving 9 of 10 correct. |

ABCD Checklist

- | | | |
|----------|-------------------|--|
| A | Audience: | <i>'Second-year pharmacy students'</i> |
| B | Behavior: | <i>'will correctly calculate dosage'</i> |
| C | Condition: | <i>'given a practical exam...'</i> |
| D | Degree: | <i>'achieving 9 out of 10'</i> |

SMART Checklist

- | | | |
|----------|--------------------|---|
| S | Specific: | <i>'Calculate dosage' — clear action verb</i> |
| M | Measurable: | <i>'9 out of 10' — quantified criterion</i> |
| A | Achievable: | <i>Appropriate for 2nd-year level</i> |
| R | Relevant: | <i>Core pharmacy competency</i> |
| T | Time-bound: | <i>'Given a practical exam' — defined event</i> |

Key Takeaways

1 Three Domains —

All learning is Cognitive (thinking), Psychomotor (doing), or Affective (feeling). Effective design addresses all three.

2 Hierarchy Matters —

Each domain progresses from lower to higher order. Build foundational objectives before advanced ones.

3 Verb Choice is Critical —

Action verbs define the observable behavior and the domain level. 'Design' signals Create (L6); 'List' signals Remember (L1).

4 Goals ≠ Objectives —

Goals are broad ideals; objectives are specific, observable, measurable statements of learner behavior.

5 ABCD Model —

Every strong objective has: Audience, Behavior, Condition, and Degree. Missing any component weakens it.

6 SMART + ABCD = Excellence —

ABCD provides structure; SMART ensures quality. Apply both frameworks to every

